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Unlocking Second Language Novel Metaphor Processing: Behavioral and ERP Insights From First and Second-Language English Users

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ABSTRACT

A growing literature uses event-related potentials (ERPs) to investigate novel metaphor processing as a window into creative processes like conceptual expansion. Modulations of the N400 generally indicate that while novel metaphors are initially processed as semantic anomalies, after a connection is found relating the concepts, they pattern more with literal sentences. Existing research largely focuses on monolinguals, but less is known about novel metaphor processing in bilinguals' second language (L2). Here, we combine robust single-trial ERPs and behavioral measures to investigate how L2 English users process full-sentence novel metaphors. We compare our results to a previous study with English monolinguals using the same experimental design to test three competing hypotheses: L2 conceptual expansion will be more effortful than, more efficient than, or similar to L1. Group differences suggest more effortful processing for L2 English users than monolinguals. Behaviorally, L2 users show more sentence evaluation errors than monolinguals, particularly for anomalous sentences. ERP results in L2 users reveal an N400 semantic anomaly effect at the sentence-final position, with no significant differences between metaphorical and literal or metaphorical and anomalous sentences. Monolinguals show a graded N400 effect, with significant differences between literal and anomalous as well as metaphorical and anomalous sentences. By comparing L2 users' results with monolingual English users and using naturalistic full-sentence structures, our findings contribute to the emergent literature on L2 novel metaphor processing and conceptual expansion while also unraveling the cognitive challenges associated with incremental processing and integration of L2 metaphorical sentences.

1 | Introduction

Creativity is commonly defined as the ability to generate ideas that are both novel (unusual or unfamiliar) as well as appropriate (useful or relevant) (Sternberg 1999). Successful communication of creative ideas is crucial for the development of innovative ideas that advance science and society. Figurative language is

one major vehicle through which individuals express creative thoughts and is a powerful tool for communicating concepts and abstract ideas (Cacciari and Glucksberg 1994). Metaphor is a particular type of creative language that acts like a bridge connecting distant and often unrelated concepts to communicate complex ideas or expressions that would otherwise be uncommunicable (Bowdle and Gentner 2005; Cacciari and

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Glucksberg 1994). Metaphors can be further divided into conventional metaphors (e.g., “Time is running out”) and novel metaphors (e.g., “The earthquake inhaled the city”; Goldstein et al. 2012; Lai et al. 2009), and the latter are the focus of the present study.

Novel metaphors exemplify both qualities of creative thought—they are unusual and unfamiliar, while also being useful and appropriately conveying meaning—thus providing a unique window into understanding creative cognition. Specifically, processing of novel metaphorical language involves a key creative process known as “conceptual expansion”—the ability to extend one’s understanding of a concept to include new characteristics and associations (Abraham et al. 2021; Rutter et al. 2012; Ward 1994). To comprehend novel metaphorical language, a listener or reader must use conceptual expansion to extend their understanding of the words in the metaphor to make new connections and access the metaphorical meaning of the phrase.

Other theories of novel metaphor processing also often emphasize the need to make connections between concepts (for reviews, see Holyoak and Stamenkovic 2018; McGregor et al. 2019). For example, the career of metaphor model (Bowdle and Gentner 2005) draws on the structural-mapping theory (Gentner 1983) and suggests that novel metaphors are comprehended like analogies and are understood via a comparison process in which features of the target concept in the metaphor are aligned with the base concept. Similarly, the conceptual blending theory (or conceptual integration theory, Fauconnier and Turner 1998) suggests that metaphor comprehension requires mapping of distantly related items as well as the activation of background knowledge (Coulson and van Petten 2002). Relevance theory argues that using metaphors entails adjusting the meaning of a lexically encoded sense and broadening its meaning to create an ad hoc concept (or occasion specific sense) based on interactions among encoded concepts, contextual information, and pragmatic expectations (e.g., Wilson and Carston 2007).¹

1.1 | Novel Metaphor Processing: Electrophysiological Studies

Electrophysiological (EEG/event-related potential [ERP]) methodologies have been particularly useful in testing these theories and examining the cognitive and neural mechanisms underlying novel metaphor processing (for recent reviews, see Canal and Bambini 2023; Citron 2023). EEG is a useful tool for directly assessing neural activity during metaphor processing and provides unique insights into the neural correlates of real-time language processing. Moreover, EEG has been found to be more sensitive than behavioral measures alone, picking up on specific and subtle effects of lexical-semantic access and conceptual expansion (cf. Jończyk et al. 2020).

Two main ERP components have been associated with novel metaphor processing: the N400 and later post-N400 components. The N400 is a negative-going waveform with a peak around 400 ms after a stimulus is presented; it was first identified as a response to semantic incongruence and is generally considered to reflect semantic access (Federmeier 2021; Kutas

and Federmeier 2011) and automatic spreading of activation in semantic memory (e.g., Kiefer 2002). The most common ERP finding is that metaphorical word pairs or phrases (e.g., *to harvest courage; love is blind*) elicit a larger (i.e., more negative) N400 amplitude than literal phrases, but a smaller or sometimes comparable N400 amplitude when compared with semantically anomalous phrases (e.g., Arzouan et al. 2007; Jankowiak et al. 2021; Lai et al. 2009; Rataj et al. 2018; Tartter et al. 2002; Wang and Jankowiak 2021; for a review, see Canal and Bambini 2023). Such a graded N400 modulation has also been observed in studies comparing novel metaphorical sentences with literal and semantically anomalous sentences (e.g., Abraham et al. 2021; Coulson and van Petten 2002; Jończyk et al. 2020; Rutter et al. 2012), suggesting that novel metaphors are more difficult to process than literal stimuli, but less or as difficult as semantically anomalous stimuli. Metaphor processing theories account for this increased processing cost as reflecting greater effort in making connections between the concepts in the metaphor. For example, Arzouan et al. (2007) interpreted their graded N400 amplitudes in terms of the conceptual blending theory, suggesting that while literal and novel metaphorical phrases are both processed by creating mappings between the concepts within them, this mapping process is more effortful for novel metaphors in which the concepts are more distantly related.

Some studies also observed amplitude differences in a later post-N400 time window, although the findings are less consistent than for the canonical N400 component, possibly due to differences in experimental task demands or stimulus types (see Rataj et al. 2018). Several studies found sustained negativities in which amplitudes for novel metaphors remain more negative than for literal sentences in the post-N400 time window (e.g., Arzouan et al. 2007; Bambini et al. 2019; Lai et al. 2009; Rutter et al. 2012). Other studies have found an increased LPC/P600 (Late Positive Complex; a positive-going wave peaking around 600–800 ms poststimulus, associated with sentence-level integration or reanalysis; Leckey and Federmeier 2019; Tanner et al. 2017) for novel metaphors compared to literal stimuli (e.g., Coulson and van Petten 2002; or familiar metaphors, e.g., Weiland et al. 2014; De Grauwe et al. 2010). Both positive and negative post-N400 components have been interpreted to represent continued efforts to integrate novel metaphorical meaning. For example, Arzouan et al. (2007) took the sustained negativities for novel metaphors to reflect additional semantic integration efforts needed to access the novel metaphorical meaning. The LPC effect observed by De Grauwe et al. (2010) was interpreted to signify a continued (re)analysis of the sentence meaning, triggered by the need to resolve the conflict incurred by the initial construction of an (implausible) literal sentence-level representation and the actual metaphorical interpretation of the critical word in the sentence.

While many studies of novel metaphor processing use simple word pairs or nominal phrases, our study adds to the smaller ERP research literature that uses full-sentence metaphors, which more closely resemble metaphors as encountered in daily life, to investigate novel metaphor comprehension as a window into creative processing captured by conceptual expansion. As we will discuss in the next session, our study is the first to investigate novel metaphorical sentence processing in the second

language. Rutter et al. (2012) and Abraham et al. (2021) both compared participants' responses to sentences with literal (e.g., Die Wolken sind über die Stadt *gezogen* [The clouds have *moved* over the city]), novel metaphorical (e.g., Die Wolken haben über der Stadt *getanzt* [The clouds have *danced* over the city]), or anomalous (e.g., Die Wolken haben über der Stadt *gelesen* [The clouds have *read* over the city]) meanings. After each sentence was presented, the participants were asked whether they thought the sentence was unusual and appropriate—unusualness (novelty) and appropriateness (usefulness) being the two defining characteristics of creative thought (Abraham et al. 2021; Kröger et al. 2013; Rutter et al. 2012; Sternberg 1999). Literal sentences are highly appropriate and usual while anomalies are inappropriate and unusual. Conceptual expansion is needed specifically for novel metaphors as these are unusual but still appropriate (i.e., they still convey useful meaning). ERP results from both studies showed larger N400 amplitudes for anomalous and novel metaphorical sentences than for literal sentences, interpreted to indicate increased effort needed to make a connection between distantly related concepts in the two “unusual” conditions, novel metaphors and anomalies.

Both Rutter et al. (2012) and Abraham et al. (2021) also observed sustained negativities in the 500–900 ms time window, with semantically anomalous sentences eliciting the most negative amplitude, followed by novel metaphors and then literal sentences. This differs from the pattern observed in the N400 window, in which anomalous and novel metaphorical sentences patterned together (Rutter et al. 2012; Abraham et al. 2021). These combined results suggest that although both semantically anomalous sentences and novel metaphors are initially interpreted as semantic incongruences (as seen from enhanced N400 to metaphorical and anomalous sentences relative to literal sentences), the use of conceptual expansion allows a plausible connection to be found for novel metaphors but not for semantically anomalous sentences (as seen from reduced amplitude of metaphorical relative to anomalous sentences in the sustained negativity time window). In other words, metaphorical sentences were initially interpreted as semantic anomalies (during the N400 time window), but upon reflection deemed to be plausible and tenable semantic expansions during the post-N400 time window (unlike the semantically anomalous sentences). Taken together, these findings support the idea of a two-step conceptual expansion process involved in novel metaphor processing; first, semantic access and activation occur and sentences are evaluated for novelty (N400), followed by an integration stage in which the distant concepts in the metaphor are related and meaning is accessed (post-N400 component).

Using a similar paradigm as Rutter et al. (2012) and Abraham et al. (2021), Jończyk et al. (2020) translated their sentences (e.g., Die Wolken haben über der Stadt *getanzt*) into English (thereby moving the critical verb to mid-sentence position, here: The clouds have *danced* over the city) and presented them to English native speakers. Time-locked to the verb (the critical word mid-sentence that determined whether a sentence frame was novel metaphorical, literal, or anomalous), N400 results mirrored N400 results of Rutter et al. (2012) and Abraham et al. (2021): Novel metaphorical and anomalous sentences both elicited more negative responses than literal sentences. Time-locked to the final word of the sentence (at which point the full-sentence

context was apparent), N400 results were like those found by Rutter et al. (2012) in the sustained negativity window: Amplitudes for novel metaphors were less negative than for anomalous sentences, but still more negative than literal sentences. These findings were interpreted to align with the conceptual expansion process proposed by Rutter et al. (2012) and Abraham et al. (2021): Novel metaphors were initially processed as an anomaly (at the verb), but conceptual expansion allowed for meaning to be reconsidered as a plausible interpretation by the time the full-sentence context was apparent (at the sentence-final word).

1.2 | Novel Metaphor Processing in the Second Language: Electrophysiological Studies

We now turn to the key topic of this study, namely the processing of novel metaphors in one's second language. The majority of electrophysiological research on novel metaphor processing has focused on monolingual individuals, leaving out most of the world's population—those who speak two or more languages. Bilinguals, particularly those who learned a second language (L2) later in life, such as immigrants, international students, or those who speak a heritage language at home, face a unique challenge: they must use creative language in their later learned and often less proficient language. However, relatively few studies have been conducted using ERPs to examine novel metaphor processing in a second language.²

These few studies observed notable differences between novel metaphor processing in a second language (L2) and a first language (L1; Jankowiak et al. 2017; Jankowiak et al. 2021; Tang et al. 2022). Most studies used a within-subject design and compared bilinguals' processing in their L1 and L2. For example, testing highly proficient Polish-English bilinguals, Jankowiak et al. (2021) found graded N400 effects for L1 Polish with larger amplitudes for anomalous nominal phrases compared to novel metaphors and novel similes compared to literal phrases. In L2 English, however, novel metaphors, novel similes, and anomalous phrases converged, all eliciting larger N400 amplitudes than literal phrases. In the post-N400 window, a sustained negativity was seen for both languages: novel metaphors and anomalous phrases elicited larger negative amplitudes than literal phrases and novel similes. This suggests that although the initial processing of novel metaphors may have been more effortful in L2, by the later time window, novel metaphors were processed similarly in both languages. However, in an earlier study using word pairs, Jankowiak et al. (2017) found a graded N400 response to anomalous, novel metaphorical, conventional metaphorical, and literal word pairs, in both participants' L1 Polish and their L2 English, indicating similar processing in L1 and L2. Still, differences were seen in peak latency, with L2 English eliciting a 20-ms delay compared to L1 Polish, and amplitude differences, with L2 English eliciting overall smaller amplitudes compared to L1 Polish. The authors suggest that the attenuated N400 amplitude for L2 English relative to L1 Polish may be due to a weaker interconnectivity of L2 semantic networks compared to L1 networks, evoking a decreased activity in memory when processing in the L2. Opposing the attenuated N400 amplitude for L2 versus L1 observed in the Polish-English bilinguals, Tang et al.'s (2022) Chinese-English bilinguals

demonstrated a larger N400 amplitude for L2 English relative to L1 Chinese novel (scientific) metaphors, and this enhanced negativity for L2 novel (scientific) metaphors was sustained in the post-N400 time window.

Additionally, proficiency may influence L2 novel metaphor processing. Wang and Jankowiak et al. (2021) found that Chinese-English bilingual participants who spoke L2 English at either an intermediate or an advanced level showed a gradual modulation of the N400 for L2 English word pairs, with the largest N400 amplitude elicited by semantically anomalous word pairs, followed by novel metaphoric, then conventional metaphoric, and then literal word pairs. However, the intermediate proficiency group elicited an increased N400 response across all word pairs compared to the advanced group, and irrespective of word pair type, which was interpreted to mean that L2 semantic memory activation was more effortful for intermediate than for proficient L2 speakers.

In sum, the few studies that examined metaphor processing in the L2 have largely yielded inconclusive evidence, but they do indicate that processing novel metaphors in the L2 differs from that in L1, possibly related to differences in the interconnectivity of L2 semantic networks compared to L1 semantic networks and in navigating novel meaning within the L2 and L1 networks. However, the research on L2 novel metaphor processing is still in its infancy compared to L1 metaphor processing, which has been examined in a large variety of contexts. For example, the L2 novel metaphor studies discussed above have used either word pairs or nominal stimuli and thus provide little information about L2 processing of more naturalistic novel metaphors such as full-sentence verbal metaphors (which are more often encountered in daily life; King and Gentner 2022). Furthermore, most studies used within-subject designs. Such designs have the advantage of comparing L1 and L2 performance within the same individual, but at the same time do not control for the abundant presence of cross-language activation in bilingual memory (for a recent review, see Van Hell 2023) while also entailing a comparison of two structurally different languages (e.g., L1 Chinese/Polish with L2 English). The present study presented full metaphorical sentences in L2 English to Chinese-English bilinguals and compared their neural responses to L2 English sentences with those of L1 English speakers presented with the same materials (reported in our earlier work, Jończyk et al. 2020). Before presenting our study in more detail, we first discuss theoretical perspectives on differences between L2 and L1 semantic memory networks.

As reviewed above, many theories on novel metaphor comprehension, including the notion of conceptual expansion, emphasize the need to expand the meaning of concepts and make new connections between concepts in semantic memory, which also pertains to processing novel metaphors in L2. There are two key perspectives on how the structure of semantic memory and conceptual expansion may differ for L2 relative to L1. On the one hand, L2 semantic networks may be more weakly linked together than L1 semantic networks because bilinguals use each language less frequently than someone who speaks only one language (cf. weaker links hypothesis, e.g., Gollan et al. 2008; Wilks and Meara 2002). Weakly linked semantic networks make connecting distant ideas more effortful, thus hindering novel

metaphor comprehension in the L2. Conceptual expansion for bilinguals performing in their L2 will thus be more effortful than for L1 users of that language.

On the other hand, the structure of L2 conceptual knowledge and bilinguals' experience navigating conceptual representations in two languages may allow for more efficient conceptual expansion. Highly creative individuals were found to have conceptual knowledge networks with flat hierarchical structure, and more interconnected (and less modular than those for low-creative individuals Kenett et al. 2014, 2016), allowing for efficient connections to be made between distantly related concepts (associative theory of creativity, Mednick 1962; Kenett et al. 2014), and facilitating conceptual expansion in online novel metaphor comprehension (Abraham et al. 2021). L2 conceptual knowledge structures have been found to have similar properties as those of highly creative individuals (Borodkin et al. 2016; but see Lange et al. 2020), and bilinguals immersed in their second language were found to have more interconnected and less modular (more efficient) L2 networks than L1 networks, regardless of L2 proficiency (Chaouch-Orozco and Martín-Villena 2024). Taken together, these findings imply bilinguals in their L2 may show more efficient conceptual expansion than L1 speakers.

1.3 | The Present Study

The present study employed behavioral and EEG methods to investigate the online processing of novel metaphors in a second language. While EEG data were recorded, participants completed a metaphor judgment task in their second language, English, in which they were shown novel metaphorical, literal, and anomalous sentences, and then judged the unusualness and appropriateness of each. This research expands on the emerging literature investigating L2 novel metaphor processing, by using full-sentence verbal stimuli which resemble metaphors as commonly encountered in daily conversation and which require readers to recruit resources to parse sentences incrementally, generate predictions about upcoming words based on grammatical and semantic information in the preceding context, and build up a hierarchical sentence structure—which may be particularly challenging in L2 readers with lower L2 proficiency levels. Additionally, this study expands on the emerging literature understanding novel metaphor processing as a creative process involving conceptual expansion, investigating conceptual expansion during L2 processing. Specifically, this study followed the procedures of and used the same stimuli as Jończyk et al.'s (2020) study of conceptual expansion in novel metaphor processing of English monolinguals. A reanalysis of this monolingual dataset combined with the current study of L2 English speakers allow for a comparison between processing of L1 and L2 speakers of English.

We addressed three hypotheses regarding how conceptual expansion of L2 English speakers may differ from that of monolingual English speakers. A first hypothesis is that conceptual expansion will be more effortful in the L2 than in the L1. Building on the notion that semantic memory networks are more weakly interconnected in L2 than in L1, conceptual expansion for bilinguals performing in an L2 will be more effortful than for L1 users of that language. In this case, the prediction

TABLE 1 | Examples of experimental stimuli.

	Literal	Novel metaphorical	Semantically anomalous
Example 1	The earthquake devastated the city	The earthquake inhaled the city	The earthquake defrosted the city
Example 2	The echo resounded through the cave	The echo trembled through the cave	The echo brushed through the cave

for L2 speakers is that at the verb both novel metaphorical and anomalous sentences are processed as anomalies compared to literal sentences; at the final word, this same pattern would continue, as conceptual expansion was not effective to make a plausible connection for the novel metaphorical sentences.

Second, if the structure of L2 conceptual knowledge and bilinguals' experience navigating conceptual representations in two languages allow for more efficient conceptual expansion, then bilinguals in their L2 may show more efficient conceptual expansion than L1 speakers. In this case, it would be predicted that at the verb, again, both novel metaphorical and anomalous sentences are processed as anomalies compared to literal sentences; at the final word, a group difference would show less negative amplitudes for novel metaphors for the L2 group compared to the L1 group, with novel metaphorical amplitudes patterning with literal sentences.

Third, if L2 and L1 conceptual expansion are similar, we predict that ERP patterns of L2 English speakers would mirror those of Jończyk et al.'s (2020) L1 English speakers: at the verb of the sentence, amplitudes for novel metaphorical and anomalous sentences would converge and both are processed as anomalies compared to literal sentences; at the final word of the sentence, at which point the full-sentence structure is apparent and conceptual expansion can be used to make plausible connections between the words in the sentence, a graded pattern would be seen with more negative amplitudes for anomalous compared to novel metaphorical, compared to literal sentences.

2 | Methods

2.1 | Participants

Thirty-six undergraduate or graduate students at the Pennsylvania State University who spoke Mandarin Chinese as their first language and English as their second language were recruited. Data from six participants were discarded—one due to poor accuracy on behavioral measures (only 3.9% of literal sentences rated as appropriate), one due to self-reported left-handedness, two due to equipment difficulties during EEG testing, and two due to poor data quality. The final sample consisted of data from 30 right-handed participants ($M_{\text{age}} = 20.53$ years, $\text{min} = 18$, $\text{max} = 27$; 15 females, 15 males, 0 non-binary), with normal or corrected-to-normal vision and no history of language or developmental disorders.

All participants reported speaking Mandarin-Chinese as their first language and English as a second language. All were born in Asia, apart from four participants who were born in the United

States but lived in a country where Mandarin is spoken for multiple years (one participant for 2–3 years, one for 4–5 years, and two for more than 10 years). All reported acquiring English before the age of 11 ($M_{\text{age}} = 5.90$, $SD = 2.22$, $\text{min} = 0$, $\text{max} = 11$). Eleven participants reported having lived in a country where English is spoken for less than a year, while seven lived in an English-speaking country for 1–5 years, nine for 5–9 years, and three for 10 or more years. Participants were Mandarin dominant, as self-reported proficiency scores (out of 10) were higher for Mandarin ($M = 9.4$, $\text{min} = 6.75$, $\text{max} = 10$) than for English ($M = 7.7$, $\text{min} = 5.25$, $\text{max} = 10$) for all but one participant. Scores on the Lexical Test for Advanced Learning of English (LexTALE; Lemhöfer and Broersma 2012; see Procedure) were 64.66 on average ($SD = 16.28$, $\text{min} = 42.88$, $\text{max} = 100$). LexTALE scores were not significantly correlated with age of acquisition ($r = 0.06$, $p = 0.77$).

2.2 | Stimulus Materials

Stimuli were identical to those used by Jończyk et al. (2020) and include 76 triads of English sentences—76 literal, 76 novel metaphorical, and 76 semantically anomalous sentences. As illustrated in Table 1 (full list in S1), sentence triads were identical except for the verb in the middle of the sentence, which changed based on condition. These critical verbs were matched for length and frequency of occurrence in the English language (see Jończyk et al. 2020). Behavioral findings from Jończyk et al. (2020) showed that native English speakers perceived the anomalous sentences as more unusual than the novel metaphorical sentences, and both of these as more unusual than the literal sentences. Furthermore, they perceived the literal sentences as more appropriate than the novel metaphorical or anomalous sentences, while the novel metaphorical sentences were more appropriate than the anomalous sentences.

To assess the appropriateness and unusualness of the English sentence materials for L1 Mandarin L2 English users, additional rating studies were conducted online via Qualtrics, following the procedures of Jończyk et al. (2020). A separate group of 36 Mandarin-English bilinguals participated in the rating study, drawn from the same population but all different from the individuals who participated in the experiment. Data from 11 individuals were excluded (nine who did not list Mandarin as their first language and two who completed the questionnaire in under six minutes). The final sample included 25 participants ($M_{\text{age}} = 19.88$, $\text{min} = 18$, $\text{max} = 22$, 7 females, 18 males) who rated both the unusualness and appropriateness of 76 novel metaphorical and literal sentences on a 4-point scale (1 = highly usual to 4 = highly unusual; 1 = highly inappropriate to 4 = highly appropriate). Participants were shown a sentence and then asked to

rate it for both unusualness and appropriateness before advancing to the next sentence. Sentences were randomized such that each participant only saw one version of each sentence (either the novel metaphorical or literal version). A t test was used to compare average unusualness and appropriateness ratings for different sentence types. There was a significant effect of sentence type on unusualness ratings ($t(150) = 7.32, p < 0.001$), confirming that novel metaphorical sentences ($M = 2.50$, 95% CI [2.43, 2.57]) were rated as more unusual than literal sentences ($M = 2.10$, 95% CI [2.02, 2.18]), as seen in Figure 1 (left panel). There was also a significant effect of sentence type on appropriateness ratings ($t(150) = 7.4, p < 0.001$), confirming that literal sentences ($M = 3.02$, 95% CI [2.96, 3.1]) were rated as significantly more appropriate than novel metaphors ($M = 2.67$, 95% CI [2.60, 2.74]), as seen in Figure 1 (right panel).

2.3 | Procedure

After providing informed consent, participants completed a Language History Questionnaire (adapted from Li et al. 2020) and the Edinburgh Handedness Questionnaire (Oldfield 1971). Participants were then seated in a sound-attenuated, dimly lit booth, approximately 100 cm from a computer screen. The experimental procedure followed that of Jończyk et al. (2020). Sentences were presented on the screen, one word at a time, while EEG was recorded. Each trial began with the presentation of a fixation cross for 500 ms, followed by each word of the sentence for 300 ms each with an inter-stimulus interval of 250 ms. At the end of each sentence, participants were asked to judge whether each sentence was unusual and appropriate. After seeing the prompt “Unusual?” on the screen they were instructed to press “yes” if the sentence seemed unusual or unfamiliar to them, and “no” if the sentence seemed usual or familiar. They then saw a prompt “Appropriate?” and pressed “yes” if the sentence seemed appropriate or made sense to them, and “no” if the sentence seemed inappropriate or did not make sense. The task was self-paced—after responding to the two questions, participants could continue to the next trial when ready. Participants first completed a practice block before completing three blocks of experimental trials. Different versions of the same sentence

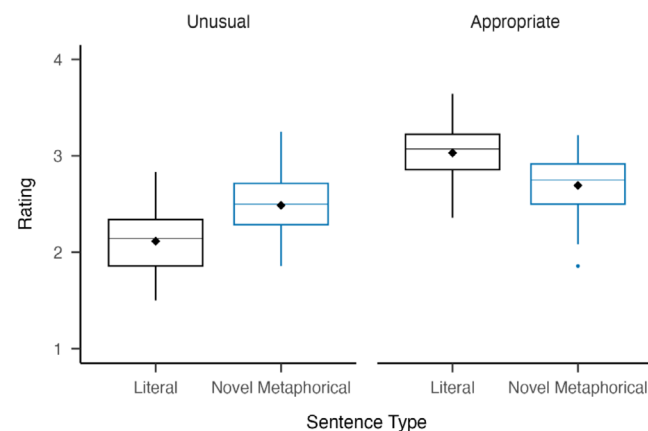


FIGURE 1 | Mean ratings for unusualness (left panel) and appropriateness (right panel) for literal and novel metaphorical sentences as rated by Mandarin-English bilinguals on a 4-point scale from 1 (highly usual/highly inappropriate) to 4 (highly unusual/highly appropriate).

frame did not appear within the same block. Block order was counterbalanced across participants and trials were randomized within blocks.

Following the experiment, participants completed multiple behavioral tasks including the Lexical Test for Advanced Learning of English (LexTALE; Lemhöfer and Broersma 2012) to measure English language proficiency. Other tasks administered included letter and semantic verbal fluency tasks in English and Chinese, a fluid intelligence test, and the Alternate Uses Task (Guilford 1967); these data were collected for the purpose of a different study.

2.4 | EEG Acquisition and Preprocessing

EEG data were recorded continuously using 30 Ag/AgCl active ActiCAP electrodes (Brain Products GmbH, Germany) placed using the extended 10–20 convention (Jasper 1958; Fp1, Fp2, F7, F3, Fz, F4, F8, FC5, FC1, FC2, FC6, T7, C3, Cz, C4, T8, CP5, CP1, CP2, CP6, P7, P3, Pz, P4, P8, PO9, O1, Oz, O2, PO10) at a sampling rate of 500 Hz and referenced to electrode FCz. Additional electrodes were placed on the left and right mastoids, as well as above and below the left eye to measure vertical eye movement, and at the outer canthus of each eye to measure horizontal eye movement. Impedances were kept below 10 kΩ. Neuroscan SynAmps2 (El Paso, TX) was used to amplify the signal, which was filtered online with a band-pass filter between 0.05 and 200 Hz.

EEG data were preprocessed and analyzed in Matlab R2023B (The MathWorks Inc) using the EEGLAB toolbox (v2024.0, Delorme and Makeig 2004). EEG data were resampled to 250 Hz. A high-pass filter at 0.1 Hz was then applied to raw EEG data using a zero-phase noncausal Hamming-windowed sinc Finite Impulse Response filter (filter order 16501, cutoff frequencies: 0.05 Hz). The sinusoidal 60-Hz line noise was estimated and removed using the *cleanline* EEGLAB plugin (v2.0, Bokil et al. 2010). Artifactual channels were identified using the *clean_rawdata* plugin in EEGLAB (correlation threshold: 0.75; line noise criterion: 4 SDs) and then removed ($M_{\text{removed}} = 1.17$; $SD = 1.04$; $\min = 0$, $\max = 4$). Independent component analysis was performed using Infomax (*Picard* plugin v1.0; maximum iteration: 500; Ablin et al. 2018; Delorme 2023) in EEGLAB. ICLabel was then applied to detect artifacts with thresholds of a probability of 0.7 for belonging to the eye component category and a probability of 0.9 for belonging to the bad channel category ($M_{\text{rejected}} = 2.37$, $SD = 0.80$; $\min = 1$, $\max = 4$; Pion-Tonachini et al. 2019). Next, *clean_rawdata* was applied once more, this time to reject bad portions of continuous data that exceeded 100 standard deviations from the clean model of the data. Finally, the previously removed bad channels were interpolated using the spherical method in EEGLAB.

After preprocessing steps were complete, data were separated into two epochs, one time-locked to the verb and one to the final word of the sentence. Verb epochs were from −200 to 600 ms (a longer epoch was not possible due to the presentation of the following word 550 ms after the verb, though it was extended to 600 ms for graphical purposes) and final word epochs from −200 to 800 ms. Recall that there was a maximum of 228 experimental

TABLE 2 | Number of accepted epochs (out of 76) per condition following artifact rejection procedures.

Sentence type	Sentence verb			Sentence final		
	Mean (SD)	Min	Max	Mean (SD)	Min	Max
Anomalous	74.60 (2.51)	67	76	74.93 (1.89)	69	76
Literal	74.67 (1.97)	69	76	74.17 (1.56)	70	75
Novel metaphorical	74.37 (2.43)	68	76	73.97 (1.77)	68	75

trials in the experiment (76 per condition). The number of accepted epochs per condition following artifact rejection procedures is listed in Table 2. We created STUDY structure in EEGLAB separately for the verb epochs and sentence final epochs. For each STUDY, we computed event-related potentials with baseline normalized to pre-stimulus activity (−200 to 0).

2.5 | ERP Analysis

ERP analyses were carried out using a hierarchical linear model approach in the LIMO EEG toolbox (Pernet et al. 2011) that deals with both within-subject variance (i.e., single trial analyses) and between-subjects variance. At the first level, a general linear model was set up for the factor Sentence Type (literal, metaphorical, anomalous), processing each subject's single trials automatically (Bellec et al. 2012) and separately for the verb in the sentence (time window 0–600 ms) and final word in the sentence (time window 0–800 ms). Parameter estimates were obtained using trial-based weighted least squares (Pernet et al. 2022). At the second level, the effect of Sentence Type was examined channel by channel with a robust one-way repeated measures ANOVA testing for differences across subjects at each time frame, separately for the verb- and word-final positions in the sentence. Reported results are corrected for multiple testing with cluster-based correction using threshold free cluster enhancement (TFCE), which has been shown to control well the family-wise error rate (Pernet et al. 2015; Smith and Nichols 2009). Post hoc comparisons were performed using paired-sample *t* tests in LIMO, likewise corrected for multiple comparisons using TFCE at an alpha level of 5%. The effect sizes for paired-sample *t* tests were computed using *limo_get_effect_size* function and correspond to Cohen's *d* ($d = 0.2$, small; $d = 0.5$, medium; $d = 0.8$, large).

2.6 | Behavioral Analysis

Behavioral data were analyzed in R (Version 4.4.1) using the *lme4* package (Version 1.1.35, Bates et al. 2015). Trials with responses subject to technical errors were excluded from behavioral analyses. Logistic mixed-effects regression models were used to analyze trial-level usualness and appropriateness decisions (1 = “yes”, 0 = “no”) by Sentence Type (literal, semantically anomalous, and novel metaphorical).³ The fixed effect of Sentence Type was coded using Helmert contrasts such that the first set of contrasts compared metaphors and anomalies and the second set of contrasts compared literal sentences to the average of metaphors and anomalies (as in Jończyk et al. 2020). It was predicted that for the “Unusual?” question, anomalous

sentences would be rated as more unusual than novel metaphors, which in turn would be rated as more unusual than literal sentences. For the “Appropriate?” question, it was predicted that literal sentences would be rated as more appropriate than novel metaphors, which in turn would be rated as more appropriate than anomalous sentences. Random intercepts for item and subject were included, in addition to random by-subject slopes for sentence type. The final models both included the full random-effect structure (Barr et al. 2013):

1. Appropriateness ~ Sentence Type + (1 + Sentence Type | Subject) + (1 | Item)
2. Unusualness ~ Sentence Type + (1 + Sentence Type | Subject) + (1 | Item)

3 | Results

3.1 | Behavioral Results

Responses to the “Unusual?” and “Appropriate?” questions were analyzed separately. For the “Unusual?” question, a significant effect of Sentence Type indicated that the proportion of anomalous sentences rated as unusual ($EMM = 0.70$, 95% CI [0.58, 0.79]) was greater than the proportion of novel metaphorical sentences rated as unusual ($EMM = 0.52$, 95% CI [0.41, 0.63]; $\beta = -0.73$, $SE = 0.15$, $z = -4.99$, $p < 0.001$). The proportion of metaphorical and anomalous sentences rated as unusual was greater than the proportion of literal sentences rated as unusual ($EMM = 0.21$, 95% CI [0.15, 0.28]; $\beta = -2.39$, $SE = 0.31$, $z = -7.751$, $p < 0.001$).

For the “Appropriate?” question, a significant effect of Sentence Type indicated that the proportion of novel metaphors rated as appropriate ($EMM = 0.72$, 95% CI [0.66, 0.78]) was greater than the proportion of anomalous sentences rated as appropriate ($EMM = 0.55$, 95% CI [0.46, 0.63]; $\beta = 0.75$, $SE = 0.16$, $z = 4.81$, $p < 0.001$). The proportion of literal sentences rated as appropriate ($EMM = 0.87$, 95% CI [0.82, 0.91]) was greater than the proportion of novel metaphors and anomalous sentences rated as appropriate ($\beta = 1.76$, $SE = 0.22$, $z = 8.00$, $p < 0.001$).

3.1.1 | Follow-Up Analyses: Comparison to English Monolingual Group

Further statistical analyses were conducted to compare the patterns of results for the English L2 group from the present study with the English monolingual group who had been tested using the same methods, materials, and procedures by Jończyk

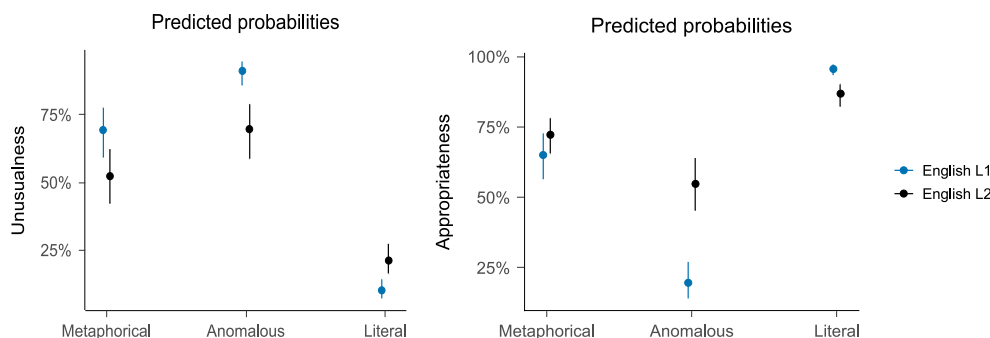


FIGURE 2 | Interaction plot depicting predicted probabilities of “yes” responses to the “Unusual?” and “Appropriate?” questions with 95% confidence intervals for each Sentence Type (literal, novel metaphorical, and anomalous) within the English L2 group and English monolingual group (data from Jończyk et al. 2020).

et al. (2020). Logistic mixed-effects regression models analyzed the binary appropriateness and unusualness decisions as a function of Sentence Type (using Helmert contrast coding, as above) and Group (using sum contrast coding, as above) as fixed effects. Item and Subject were included as random intercepts, with random by-subject slopes for Group and random by-item slopes for Sentence Type. Only significant effects of the factor Group will be reported. The final models both supported the full random-effect structure (Barr et al. 2013):

1. Appropriateness ~ Sentence Type * Group + (1 + Sentence Type | Subject) + (1 + Group | Item)
2. Unusualness ~ Sentence Type * Group + (1 + Sentence Type | Subject) + (1 + Group | Item)

For the “Unusual?” question, there was a significant interaction between Sentence Type and Group, as seen in Figure 2. English monolinguals rated anomalous sentences as more unusual than metaphorical sentences, relative to L2 English users ($\beta = -0.78$, $SE = 0.21$, $z = -3.70$, $p < 0.001$). Similarly, English monolinguals rated anomalous and metaphorical sentences as more unusual than literal sentences compared to L2 English users ($\beta = -2.64$, $SE = 0.39$, $z = -6.70$, $p < 0.001$).

The interaction between Sentence Type and Group was again significant for the “Appropriate?” question, as seen in Figure 2. English monolinguals rated metaphorical sentences as more appropriate than anomalous sentences relative to L2 English users ($\beta = 1.26$, $SE = 0.21$, $z = 5.90$, $p < 0.001$). Similarly, English monolinguals rated literal sentences as more appropriate than metaphorical and anomalous sentences relative to L2 English users ($\beta = 2.88$, $SE = 0.37$, $z = 7.88$, $p < 0.001$).

3.2 | ERP Results

At the verb of the sentence, the main effect of Sentence Type was significant in a region between 476 and 524 ms ($F_{\min} = 3.42$, $F_{\text{mean}} = 6.54$, $F_{\max} = 13.32$), covering channels: FC1, FC2, C3, Cz, C4, CP5, CP1, CP2, P7, P3, Pz, PO9, O1, Oz, and O2, as shown in Figure 3a–d. A TFCE-corrected paired t test comparing literal and semantically anomalous conditions revealed a significant region between 496 and 504 ms ($t_{\min} = 3.5$, $t_{\text{mean}} = 4.2$, $t_{\max} = 5.53$), with large effect sizes ($d_{\min} = 0.83$, $d_{\text{mean}} = 0.99$,

$d_{\max} = 1.30$; maximal effect at 500 ms at Pz), covering channels: Cz, CP1, CP2, and Pz. Courseplots for this and all following TFCE-corrected paired t test comparisons can be found in Supporting Information on OSF.⁴ Other comparisons were not significant.

At the final word of the sentence, the main effect of Sentence Type was significant in a region between 396 and 468 ms ($F_{\min} = 3.36$; $F_{\text{mean}} = 5.62$; $F_{\max} = 12.83$), covering channels: Fp1, Fp2, F7, F3, Fz, F4, FC5, FC1, FC2, FC6, T7, C3, Cz, C4, T8, CP5, CP2, CP6, P4, and P8, as shown in Figure 3e–h. A TFCE-corrected paired t test comparing literal and semantically anomalous sentences revealed two significant regions. First, significant differences emerged between 112 and 176 ms ($t_{\min} = -4.61$, $t_{\text{mean}} = -2.65$, $t_{\max} = -1.85$), with medium effect sizes ($d_{\min} = 0.44$, $d_{\text{mean}} = 0.63$, $d_{\max} = 1.08$; maximal effect at 148 ms at channel CP1), covering channels Fp2, Fz, FC1, FC2, FC6, C3, Cz, C4, T8, CP5, CP1, CP2, CP6, P7, P3, Pz, P4, P8, PO9, O1, Oz, O2, PO10. Critically, a second region corresponding to the N400 effect showed significant differences starting at 324 ms and ending at 572 ms ($t_{\min} = 1.38$, $t_{\text{mean}} = 2.36$, $t_{\max} = 6.33$), with medium effect sizes ($d_{\min} = 0.33$, $d_{\text{mean}} = 0.56$, $d_{\max} = 1.49$; maximal effect at 468 ms at channel FC6), covering channels Fp1, Fp2, F7, F3, Fz, F4, F8, FC5, FC1, FC2, FC6, T7, C3, Cz, C4, T8, CP5, CP1, CP2, CP6, P7, P3, Pz, P4, P8, PO9, O1, Oz, O2, PO10. The other comparisons were not significant ($p_s > 0.05$).

3.2.1 | Follow-Up Analyses: Comparison of L2 English Speakers With English Monolingual Speakers

For sake of comparison with L2 user data presented above, the monolingual data from Jończyk et al. (2020) were preprocessed here with the exact same pipeline used for the L2 data and likewise analyzed with LIMO EEG (preprocessing pipeline and EEG descriptive statistics including rejected channels, epochs, and ICs are available on OSF). Course plots and ERPs for the verb and final word of the sentence for the monolingual group can be seen in Figure 4.

To compare the data of the L2 English speakers with the English monolingual group ($N = 35$) from Jończyk et al. (2020), we fit a robust repeated measures ANOVA in LIMO EEG to test for the possible interaction between Sentence Type (literal, novel

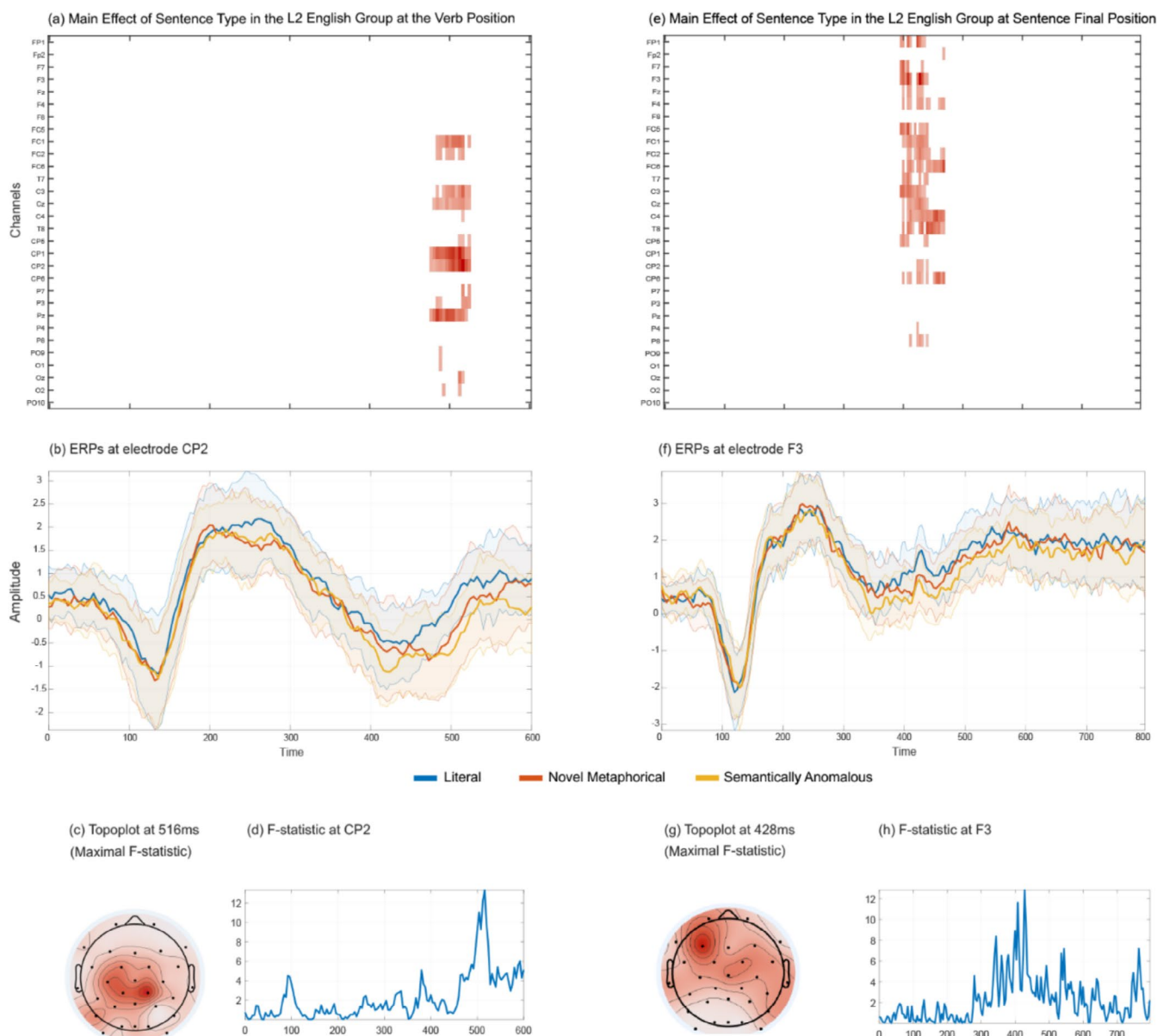


FIGURE 3 | (a) TFCE regional effects at 5% alpha level for the main effect of sentence type in the L2 English group at the verb position; (b) Grand average ERP waveforms computed as the 20% trimmed mean of the mean subject-level single-trial ERP data. Waveforms at CP2, where the maximum F -statistic was reported. Shaded areas denote the 95% Bayesian highest density interval; (c) Topoplot at 516ms where the effect was maximal; (d) F -statistic values over time at electrode CP2 where the effect was maximal; (e) TFCE regional effects at 5% alpha level for the main effect of sentence type in the L2 English group at the sentence-final position; (f) Grand average ERP waveforms computed as the 20% trimmed mean of the mean subject-level single-trial ERP data. Waveforms at F3, where the maximum F -statistic was reported. Shaded areas denote the 95% Bayesian highest density interval; (g) Topoplot at 428ms where the effect was maximal; (h) F -statistic values over time at electrode F3 where the effect was maximal.

metaphorical, semantically anomalous) and Group (L2 English, monolingual English) at the verb and sentence-final positions.

The Sentence Type by Group interaction was significant yet restricted to the sentence-final position (see Figure 5). We observed 10 regional effects with significant differences extending beyond 10 ms and covering at least three channels. Table 3 shows the time range and channels in which we observed significant effects within each region along with corresponding F -statistics. Of particular interest are regions 4–10, since they correspond to the N400 (4–8) and post-N400 (9–10) time windows.

TFCE-corrected paired t tests did not reveal significant differences between monolingual and L2 English groups for literal, novel metaphorical, and semantically anomalous sentences; therefore, the interaction is driven by within-group differences. In the monolingual group, we found a significant regional effect for the contrast between literal and anomalous sentences and two significant regional effects for the contrast between novel metaphorical and anomalous sentences, with medium to large effect sizes in both cases, as described in Table 4. In the L2 English group, we found only significant regional effects for the contrast between literal and anomalous sentences, with medium to large effect sizes, as described in Table 5.

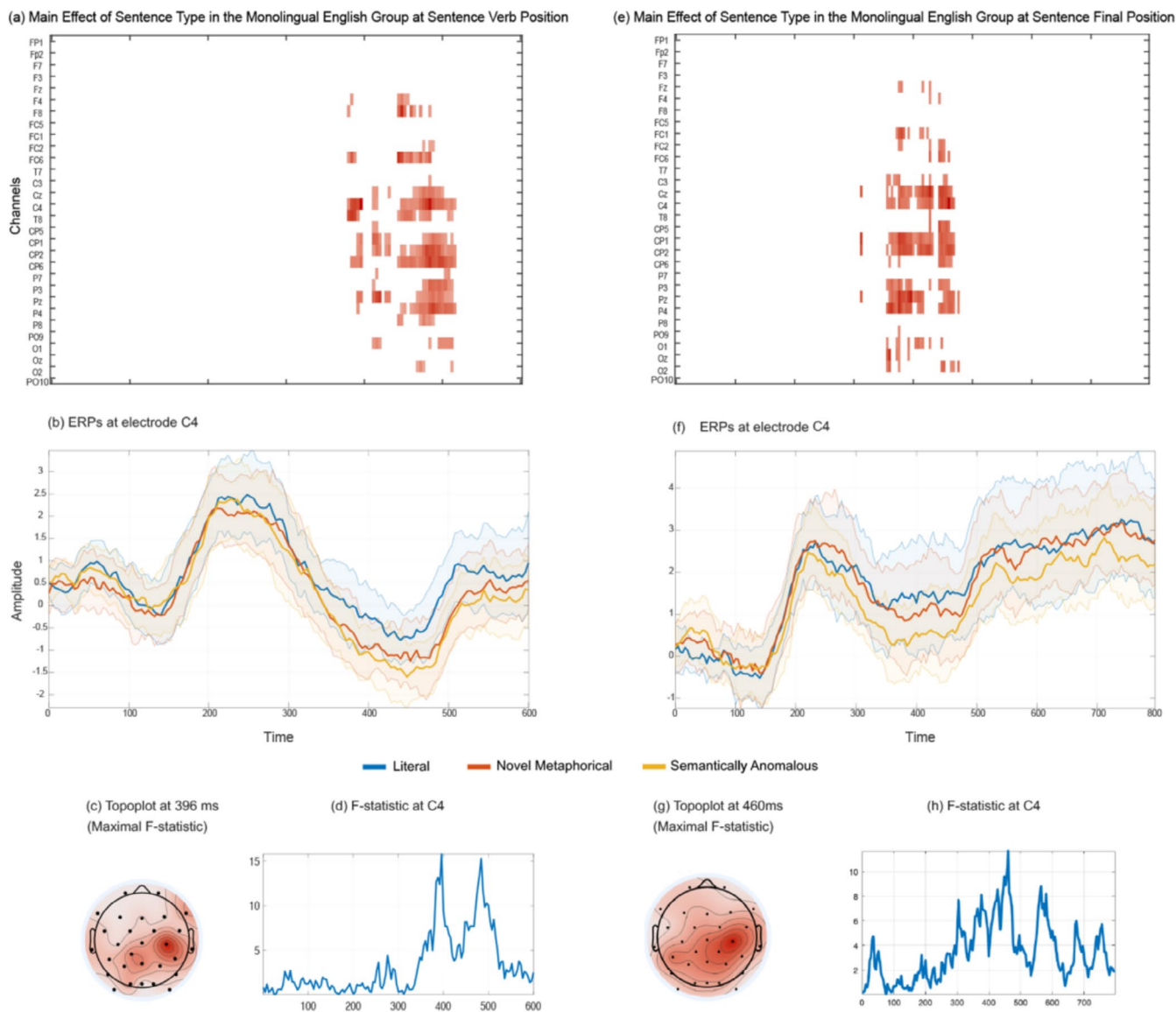


FIGURE 4 | (a) TFCE regional effects at 5% alpha level for the main effect of sentence type in the English monolingual group at the verb position; (b) Grand average ERP waveforms computed as the 20% trimmed mean of the mean subject-level single-trial ERP data. Waveforms at C4 where the maximum F -statistic was reported. Shaded areas denote the 95% Bayesian highest density interval; (c) Topoplot at 396 ms where the effect was maximal; (d) F -statistic values over time at electrode C4 where the effect was maximal; (e) TFCE regional effects at 5% alpha level for the main effect of sentence type in the English monolingual group at the sentence-final position; and (f) Grand average ERP waveforms computed as the 20% trimmed mean of the mean subject-level single-trial ERP data. Waveforms at C4 where the maximum F -statistic was reported. Shaded areas denote the 95% Bayesian highest density interval; (g) Topoplot at 460 ms where the effect was maximal; (h) F -statistic values over time at electrode C4 where the effect was maximal.

3.3 | Exploratory Analysis: Correlations With L2 Proficiency

Exploratory follow-up analyses were conducted to examine correlations between L2 English proficiency (as measured by LexTALE score) and both behavioral (average proportion of “yes” responses by participant by sentence type) and ERP results (N400 averaged across the time frame and electrodes in the significant verb and final word clusters reported above by participant by sentence type). The proportion of “yes” responses to the “unusual?” question for literal sentences decreased as proficiency

increased ($r = -0.48$, $p = 0.007$) as seen in Figure 6a. The proportions of “yes” responses to the “appropriate?” question increased for literal sentences ($r = 0.66$, $p < 0.001$) and novel metaphors ($r = 0.57$, $p < 0.001$) as proficiency increased, as seen in Figure 6b. Regarding ERP results, there were no significant correlations between LexTALE score and N400 amplitude at the final word of the sentence for any sentence type, as can be seen in Figure 6c. At the verb of the sentence, N400 amplitude was significantly negatively correlated with L2 proficiency for all three sentence types: literal ($r = -0.36$, $p = 0.049$), novel metaphorical ($r = -0.39$, $p = 0.03$), and anomalous ($r = -0.38$, $p = 0.04$), as seen in Figure 6d.

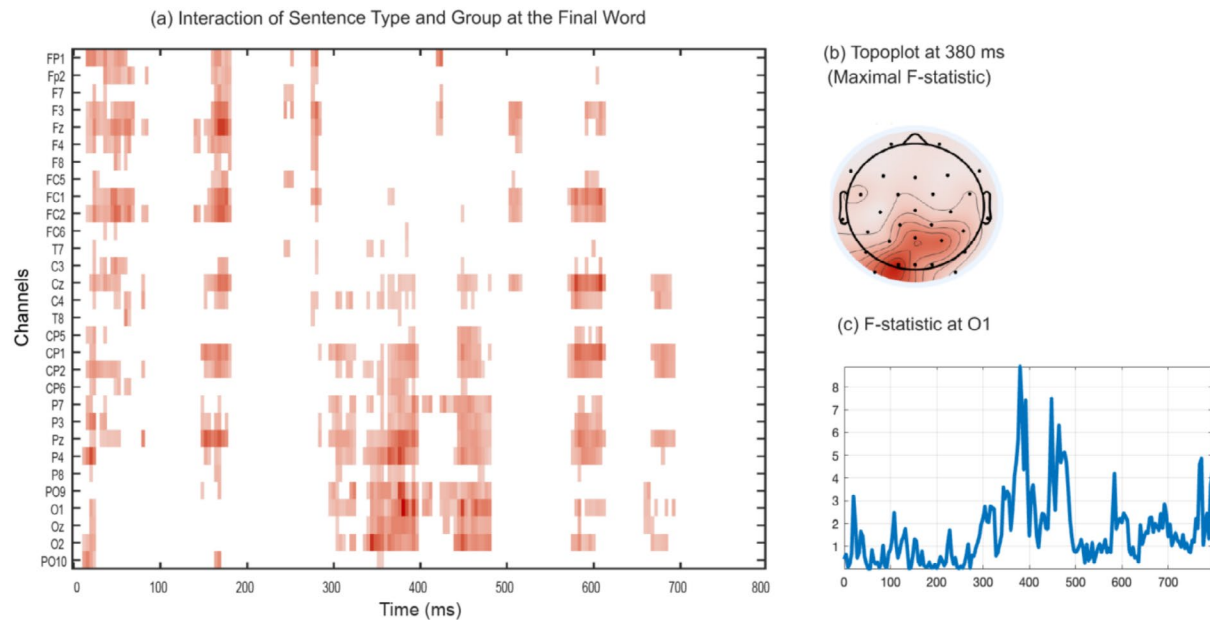


FIGURE 5 | (a) TFCE regional effects at 5% alpha level for the interaction between Sentence Type and Group at sentence-final position, (b) Topoplot at 380 ms where the effect was maximal, and (c) F -statistic power over time at electrode O1 where the effect was maximal.

TABLE 3 | Significant channels and time ranges for each regional effect of the interaction.

Region	Significant channels	Time (ms)	F_{mean}	F_{min}	F_{max}
1	Fp1, Fp2, F7, F3, Fz, F4, F8, FC5, FC1, FC2, FC6, T7, C3, Cz, C4, T8, CP5, CP1, CP2, CP6, P7, P3, Pz, P4, P8, O1, Oz, O2, PO10	12–68	2.97	1.73	7.15
2	Fp1, Fp2, F7, F3, Fz, F4, F8, FC5, FC1, FC2, C3, Cz, C4, CP1, CP2, P7, P3, Pz, P4, P8, PO9, PO10	148–180	3.52	1.83	7.72
3	Fp1, Fp2, F7, F3, Fz, F4, F8, FC5, FC1, FC2, T7, C3, C4, T8, CP1, P3, Pz	276–284	2.88	1.75	5.68
4	C4, CP1, CP2, P7, P3, Pz, P4, P8, PO9, O1, Oz, O2	296–324	2.55	1.64	4.76
5	FC1, FC6, T7, Cz, C4, T8, CP5, CP1, CP2, CP6, P7, P3, Pz, P4, P8, PO9, O1, Oz, O2, PO10	336–396	3.24	1.51	8.89
6	P7, PO9, O1	404–412	2.97	1.99	3.80
7	FC2, C3, Cz, C4, CP5, CP1, CP2, CP6, P7, P3, Pz, P4, P8, PO9, O1, Oz, O2	424–480	3.08	1.51	7.50
8	F3, Fz, F4, FC5, FC1, FC2, Cz	504–516	2.98	1.80	4.11
9	Fp2, F3, Fz, FC5, FC1, FC2, T7, C3, Cz, C4, CP5, CP1, CP2, P7, P3, Pz, P4, P8, O1, Oz, O2	572–612	3.39	1.83	7.77
10	Cz, C4, CP1, CP2, Pz, P4, PO9, O1, Oz, O2	660–692	2.76	1.92	4.03

4 | Discussion

This study aimed to investigate neurocognitive processes underlying novel metaphor processing in a second language. Much of the previous work investigating creative processes involved in novel metaphor comprehension has focused on monolingual individuals; however, the processing of novel metaphors in an L2 may differ due to unique characteristics of L2 speakers' semantic network structure. We examined L2 English speakers' behavioral and neural responses to full-sentence verbal novel metaphors (building on the emerging body of research on L2 metaphor processing, which has predominantly used word pairs or nominal metaphors) as they better mimic metaphors used in

daily language and reflect the cognitive challenges associated with conceptual expansion. Furthermore, additional analyses were conducted using data from monolingual English speakers, reported in Jończyk et al. (2020), who completed the same task using the same procedure. This between-group comparison allowed for comparisons in L2 versus monolingual processing of an identical set of novel metaphors.

Our results showed differences in the processing of full-sentence verbal novel metaphors for L2 speakers of English compared to English monolinguals. When analyzing results from just L2 English speakers, contrary to previous results on novel metaphor processing (both with first and second

TABLE 4 | Significant channels and time ranges for each regional effect of the significant contrasts in the monolingual group.

Contrast	Region	Significant channels	Time (ms)	Time					
				t_{mean}	t_{min}	t_{max}	d_{mean}	d_{min}	d_{max}
Literal—Anomalous	1	Fp1, Fp2, F7, F3, Fz, F4, F8, FC5, FC1, FC2, FC6, T7, C3, Cz, C4, T8, CP5, CP1, CP2, CP6, P7, P3, Pz, P4, P8, PO9, O1, Oz, O2	312–512	3.56	2.39	5.58	0.72	0.52	1.22
	2	FC1, FC2, C3, Cz, CP1, Pz	580–600	3.52	3.19	4.23	0.78	0.70	0.92
Metaphorical—Anomalous	1	Fp1, Fp2, F7, F3, Fz, F4, F8, FC5, FC1, FC2, FC6, C3, Cz, C4, T8, CP5, CP1, CP2, P3	404–468	3.16	2.56	4.81	0.69	0.56	1.05
	2	FC1, FC2, C3, Cz, CP1, Pz	580–600	3.52	3.19	4.23	0.78	0.70	0.92

TABLE 5 | Significant channels and time ranges for each regional effect of the contrast between literal and anomalous sentences in the bilingual group.

Region	Significant channels	Time (ms)	t_{mean}	t_{min}	t_{max}	d_{mean}	d_{min}	d_{max}
1	Fp1, F7, F3, FC5, FC1, T7, C3	408–412	3.96	3.09	5.31	0.93	0.73	1.25
2	F4, FC2, FC6, C4, T8, CP6	448–480	3.75	2.99	6.33	0.88	0.70	1.49
3	FC6, T8, CP6	488–492	3.75	3.28	4.11	0.88	0.77	0.97

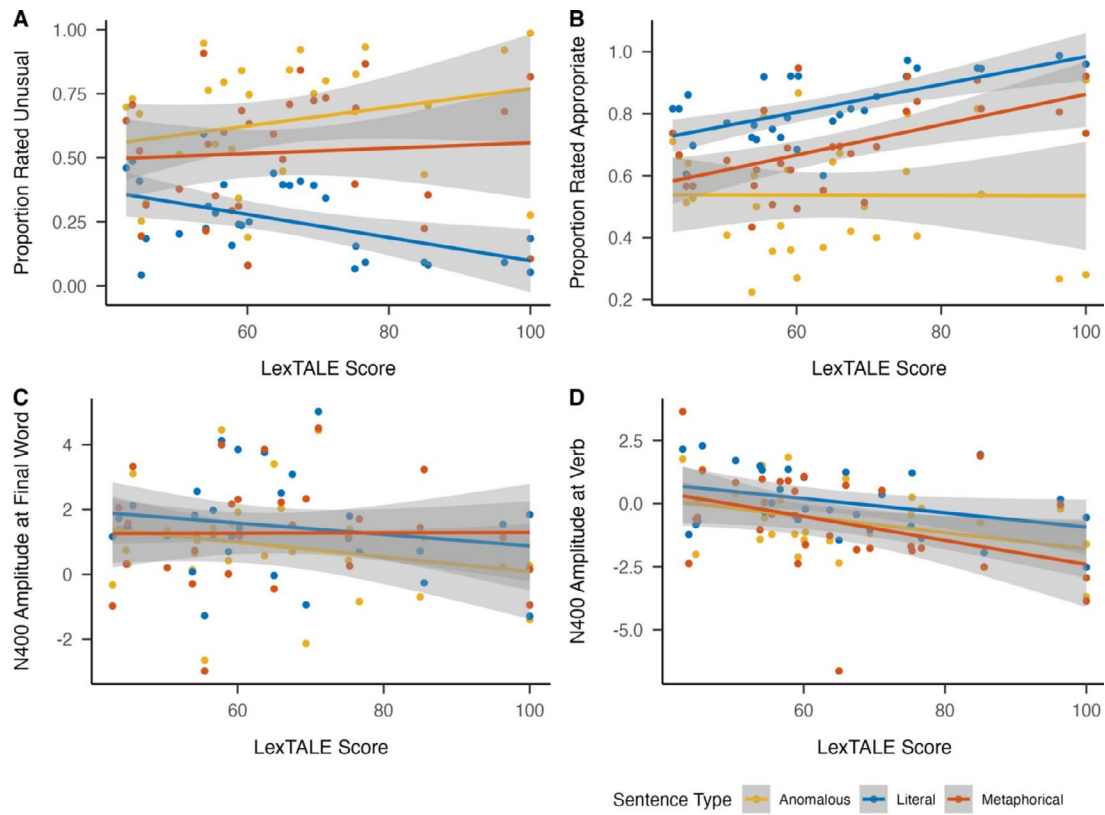


FIGURE 6 | Correlations between proficiency as measured by LexTALE score and proportion of sentences rated as unusual (a), proportion of sentences rated as appropriate (b), N400 amplitude at the final word (c), and N400 amplitude at the verb of the sentence (d) by sentence type.

language users), we did not find the typical graded N400 response to literal, novel metaphorical, and semantically anomalous stimuli at either the verb or final word of the sentence.

At the mid-sentence verb, we observed a short-lasting but significant difference in N400 amplitudes at four electrode sites for literal relative to anomalous sentences. At the sentence-final

word, we again found significant N400 amplitude differences with more negative N400 amplitudes for anomalous sentences than for literal sentences (and no effects in the post-N400 time window). As the N400 component is associated with spreading of activation in the semantic network, these results suggest that participants were sensitive to semantic anomalies in L2 sentences and that, as expected, processing meaning for anomalous sentences required greater effort than for literal sentences. However, the N400 amplitudes for the novel metaphors were not significantly different from either the literal or anomalous sentences.

A between-group comparison of the L2 speakers with monolingual English speaker data from Jończyk et al. (2020) revealed significant differences in processing between L2 English users and English monolinguals which were restricted to the final word of the sentence. While in monolinguals N400 amplitudes at the final word of the sentence indicated the greatest processing difficulty for anomalous sentences, followed by novel metaphorical and literal sentences, L2 English users differentiated only between literal and anomalous sentences, showing less sensitivity to novel metaphors. Further differences between L2 English and monolingual English speakers emerged in the timing and topography of the significant clusters for the two groups. In L2 speakers, the reported effects show a more fronto-central distribution and seem shorter-lived; in monolingual English speakers, the effects are distributed over centro-parietal regions and are more robust and extensive. Note that at the mid-sentence verb, although semantic anomaly effects for the L2 group were short-lasting, we did not find significant differences in processing between the L2 and monolingual groups.

Competing hypotheses regarding the structure of L2 semantic memory and conceptual expansion in L2 metaphor comprehension were tested in the context of full-sentence metaphors. Assuming that conceptual L2 semantic networks are more weakly interconnected than L1 semantic networks, making conceptual expansion in L2 more effortful, the weaker-links perspective predicted that at the mid-sentence verb, both novel metaphorical and anomalous sentences are processed as anomalies compared to literal sentences; this pattern was predicted to continue at the sentence-final word, as conceptual expansion was not effective to make a plausible connection for the novel metaphorical sentences. The alternative hypothesis predicted that L2 conceptual expansion would be facilitated, because of the flexible structure of bilingual semantic networks (which are found to be similar to those of highly creative individuals, Abraham et al. 2021; Borodkin et al. 2016; Kenett et al. 2018). The pattern of results we observe (only a semantic anomaly N400 effect at the mid-sentence verb and sentence-final word, and overall less robust and more attenuated N400 effects in the L2 compared to L1) more strongly supports the hypothesis that conceptual expansion is more effortful in L2 compared to L1 processing. Although at the mid-sentence verb and sentence-final word, L2 English speakers were sensitive to downright semantic violations (i.e., as N400 amplitude was significantly more negative for the anomalous than for literal sentences), the more subtle semantic manipulations (novel metaphors) did not modulate the N400 amplitude. These results indicate that, while for monolinguals, plausible connections could be found relating the concepts in the novel metaphor, conceptual expansion for

L2 speakers was more difficult. This interpretation aligns with related theories of metaphor processing, such as the conceptual blending theory (Coulson and van Petten 2002), the career of metaphor model (Bowdle and Gentner 2005), and relevance theory (Wilson and Carston 2007), which also emphasizes the need to make connections between concepts. Our results suggest that for L2 speakers, making these connections between concepts is even more effortful, particularly while processing full sentences, a topic we will turn to next.

The N400 differences between literal and anomalous sentences at the mid-sentence verb and sentence-final word (with little sensitivity to novel metaphors) for L2 users may be due to difficulties with incremental processing of full sentences in their L2. Results from Citron et al.'s (2020) fMRI study finding differences during L1 and L2 figurative language processing support this idea. L1 users showed activation differences when reading metaphorical versus literal sentences; however, activation patterns for L2 users largely did not differ between sentence types. Moreover, across sentence types, L2 users showed increased activation in areas relating to silent reading comprehension and motor planning, as well as areas in the executive control network associated with language switching and conflict monitoring. L1 users, on the other hand, showed more activation in the default mode network—associated with mind wandering or mentalizing (Citron et al. 2020) as well as creativity and novel metaphor production (Beaty et al. 2017). Citron et al. (2020) interpreted these results to mean that the task was overall less cognitively demanding for L1 users who could allocate more resources toward meaning integration than could L2 users. This supports our L2 speakers' results showing processing differences in real time only for literal versus semantically anomalous sentences at the sentence-final word, but not for novel metaphors at either the mid-sentence verb or sentence-final word, which require conceptual expansion to access meaning.

The present study's ERP measures, capturing sentence processing in real time, indicate that conceptual expansion during incremental sentence processing is more effortful for L2 English speakers than for English monolinguals. The behavioral results suggest that this difficulty may continue during offline sentence comprehension after the full sentence has been processed. On the one hand, the pattern of effects of the L2 English speakers was similar to that of the English monolinguals—for both groups, anomalous sentences were rated as more unusual than novel metaphors and both were rated as more unusual than literal sentences; literal sentences were rated as most appropriate and novel metaphors were more appropriate than anomalous sentences—which indicates eventual successful conceptual expansion in the L2. However, group differences indicate that monolinguals were, overall, significantly more accurate than L2 speakers. For example, L2 users showed a reduced behavioral sensitivity to anomalies, as they evaluated the appropriateness of anomalous sentences at a chance level. Furthermore, correlations with L2 proficiency indicate that (for literal and novel metaphorical sentences), as proficiency increases, L2 users' behavioral responses increasingly resemble that of monolinguals. High levels of inaccuracy for lower-proficiency participants as well as overall reduced sensitivity to anomalies may explain the lack of a clear graded effect in the N400 range at the sentence-final word. Relatedly, it is conceivable that participants with

lower proficiency in the L2 may be confusing anomalies with novel metaphorical sentences, particularly in a task involving novel metaphor evaluation. Indeed, behavioral results indicate that in 50% of cases, anomalous sentences were identified by L2 users as appropriate, an evaluation expected of a novel metaphorical rather than anomalous sentences. Such a contextual effect could point to a somewhat greater tolerance of anomaly and offer a complementary explanation accounting for the lack of significant differences between metaphorical and anomalous sentences in L2 speakers.

In line with previous studies of L2 novel metaphor processing (i.e., Jankowiak et al. 2017, 2021; Tang et al. 2022; Wang and Jankowiak 2021), our results indicate differences between L1 and L2 novel metaphor processing at the neural level. However, exactly how L2 novel metaphor processing differs from that of L1 processing is inconclusive—for example, one previous study found graded N400 amplitudes in L1 but not L2 (Jankowiak et al. 2021), another found graded amplitudes in both languages but delayed peak latency in L2 (Jankowiak et al. 2017), and our results show little sensitivity to L2 novel metaphors instead only finding semantic anomaly effects. Our results may differ from these previous studies based on two factors: stimulus type and participant second-language background.

One critical difference between the current study and previous L2 studies on metaphor processing is the type of metaphor stimuli presented. Previous L2 studies have used stimuli that explicitly draw attention to metaphorical connections between concepts. For example, Jankowiak et al. (2017) and Wang and Jankowiak et al. (2021) used two-word dyads (i.e., *to harvest courage*), while Jankowiak et al. (2021) used metaphors in the nominal A is B format (i.e., *love is a monastery*). Stimuli in the current experiment were longer and more naturalistic sentences in which the verb changed to alter the meaning from literal to metaphorical to anomalous (i.e., “the earthquake devastated the city” vs. “the earthquake inhaled the city” vs. “the earthquake defrosted the city”). Compared to word dyads and nominal metaphors, full-sentence stimuli in the current study are also grammatically and semantically more complex, placing increased demands on working memory. Sentences also require readers to parse the sentence incrementally, predict upcoming words based on grammatical and semantic information in the preceding sentence context, and build up the hierarchical structure of the sentence as the sentence unfolds over time. Full metaphorical sentences are also more complex in that the metaphorical meaning is embedded within the larger sentence context, and the structure itself is not stripped back to only those concepts that need to be connected to access meaning (as in “A is B phrases” used in Jankowiak et al. 2021) or the explicit mentioning of the verb and noun to be connected, as in *to harvest courage* (used in Jankowiak et al. 2017). Relatedly, the sentences used in the current study are novel verbal metaphors, which unlike nominal metaphors that relate two concrete nouns, may involve an intermediate process of connecting sensorimotor information or the physical meaning of a verb to a more abstract meaning. Previous research has indicated that the involvement of sensorimotor information in processing novel verbal metaphors may be particularly important for lower-level L2 learners who show greater embodiment effects (i.e., Feng and Zhou 2021).

Finally, differences between results from the current study and previous studies with L2 users could be driven by differences in participants' second language experience, including their L2 proficiency. Proficiency for the current participant group was lower than that of participants tested in some of the previous L2 metaphor studies (i.e., highly proficient students of English philology immersed in an instructive English context; Jankowiak et al. 2021; Jankowiak et al. 2017; L2 proficiency level of Tang et al.'s Chinese L2 speakers living in China is not reported). However, our sample is typical of intermediate English learners (i.e., Wang and Jankowiak's 2021 intermediate group which had an average LexTALE of 60.12), which represent a large amount of L2 users who must use English in their daily classes and extracurricular activities at a large American university and regularly use and understand creative or metaphorical language in their L2. Proficiency has been shown to play a role in metaphor processing. Wang and Jankowiak et al. (2021), for example, found increased N400 activation in intermediate compared to advanced L2 learners (independent of sentence type), which was interpreted to indicate less automatic processing for intermediate users. Ibáñez et al. (2010), in their study of metaphor and gesture, found no N400 differences between metaphorical and literal phrases for upper intermediate proficiency L2 German users, while for highly proficient L2 German users N400 results were the same as for native German users. The present study was conducted using novel metaphors with a full-sentence structure, which may make L2 proficiency effects even more pronounced. To investigate the potential effects of proficiency on novel metaphor processing, we conducted exploratory correlational analyses. N400 results did not correlate with proficiency at the final word of the sentence. At the verb of the sentence, N400 amplitudes were overall more negative as proficiency increased; however, this correlation did not differ by sentence type. Correlations between behavioral responses to the “unusual” and “appropriate” questions and LexTALE scores indicated that participants with greater L2 English vocabulary knowledge had greater accuracy—as LexTALE score increased, literal sentences were rated as less unusual, and both novel metaphors and literal sentences were rated as more appropriate. In these ways, sentence ratings were more closely aligned with monolingual ratings as proficiency increased. Further research, testing full-sentence metaphor processing in L2 users at varying L2 proficiencies, will be needed to further elucidate the impact of L2 proficiency on conceptual expansion in more full novel metaphorical sentences.

In conclusion, this study aimed to investigate L2 conceptual expansion by examining participants' behavioral and neural responses to literal, novel metaphorical, and semantically anomalous sentences in their L2 English. This study expands the emergent literature on the neurocognitive correlates of L2 metaphor processing by (1) comparing results with those of monolingual English speakers and (2) using complex full-sentence structures in L2 users (opposed to previous work exclusively testing word dyads or nominal metaphors), which more closely represent novel metaphors as encountered in daily conversation and better reflect the cognitive challenges associated with accessing conceptual networks during incremental sentence processing and the integration of upcoming information in the evolving sentence structure. It also expands on the work investigating novel metaphor comprehension as a creative process

involving conceptual expansion by examining how this process occurs when using a second language. ERP and behavioral group comparisons show clear differences in the processing of novel metaphorical sentences between L2 English users and English monolinguals. N400 results showing only a semantic anomaly rather than a graded effect provide evidence for more effortful conceptual expansion in L2 users in real time. Although L2 users and monolingual English speakers showed similar response patterns at the behavioral level, overall reduced accuracy and significant correlations between behavioral responses and L2 proficiency suggest continued difficulty in processing full-sentence novel metaphors offline, particularly for those with lower L2 proficiency. Future work is needed to further understand the complex dynamics of L2 novel metaphor comprehension as sentence processing unfolds in real time, including the impact of metaphorical stimuli type and aspects of the bilingual language experience, such as different L1-L2 combinations and varying levels of L2 proficiency.

Author Contributions

Adriana R. Miller: conceptualization, data curation, formal analysis, investigation, methodology, writing – original draft, writing – review and editing. **Rafał Jończyk:** formal analysis, methodology, writing – review and editing. **Holly A. Zaharchuk:** formal analysis, methodology, writing – review and editing. **Janet G. van Hell:** conceptualization, formal analysis, funding acquisition, methodology, supervision, writing – original draft, writing – review and editing.

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Ethics Statement

This study was approved by the Institutional Review Board at the Pennsylvania State University. All participants provided written informed consent and received compensation for their voluntary participation.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

MATLAB code for EEG data preprocessing, R code used for the analysis of behavioral data, stimuli, de-identified data, and other supplemental analyses and materials are available on OSF: https://osf.io/uxh39/?view_only=49caa871deab46e885855b0a51ab60d9.

Endnotes

¹ An extensive discussion of theories of metaphor processing, including theories that focus on differences between conventional and novel metaphors, such as the graded salience hypothesis (Giora 2003), is

outside the scope of this paper (for more information, see Bambini and Domaneschi 2024; Holyoak and Stamenkovic 2018).

² L2 processing of other types of figurative language, such as irony, humor, and idiom fall outside the scope of the present work; for a review on processing figurative language in L2, see Citron (2023).

³ Behavioral analyses were restricted to yes and no response accuracy—reaction times were not analyzed as participants made two judgments about each sentence and may have had different responses strategies (i.e., thinking of both answers before responding to the first question or thinking of answers as they responded sequentially).

⁴ TFCE is a conservative full scalp correction method. An alternative and less conservative approach is simple post hoc contrasts. In the Supporting Information, we report as supplementary analysis simple post hoc contrasts for the robust repeated measures ANOVA (computed in LIMO EEG) at the verb and word final positions along with effect sizes computed using Mahalanobis Distance (MD; `limo_get_effect_size` function in LIMO EEG).

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.